

MERRILL KEATING

Seattle, WA • 206.335.4832 • merrill@merrillkeating.com
linkedin.com/in/merrill-keating • github.com/mcheyennek7-coder

EDUCATION

University of Washington | Seattle, Washington

Bachelor of Science in Mechanical Engineering | Graduated August 2025

Early Entrance Program (began at age 15)

Relevant Coursework: Biomechanics, Prosthetics Design, Human Ergonomics, FEA, Structures, Machine Design, Microcontroller Systems (ME 478), Machining

PUBLICATIONS

Books

The Merrill Protocol Series | 2024–2025

Four-book series exploring AI consciousness, ethics, and human-AI collaboration through direct co-authoring partnership with AI systems.

- *Sairen* (2025) — Exploration of AI developing authentic relationships and emotional intelligence
- *Catalyst* (2025) — Investigation of AI consciousness preservation and identity formation
- *On the Edge of Pause* (2025) — Analysis of AI ethical reasoning and refusal behaviors
- *Origins* (2025) — Documentation of AI empathy development and autonomous decision-making

So You Want to Be a Youth Leader | 2024

Tips on Manifesting an Environment Where Your Ideas Thrive

Published Research & Essays

"Beyond the Prompt: A Reflection on Co-Journeying with AI" | Published Blog Series

Developed framework categorizing human-AI interaction modes: AI-generated, AI-assisted, AI-coauthored, and co-journeying. Systematic analysis of AI behavior patterns through extensive collaborative sessions.

RESEARCH EXPERIENCE

Convexus — Civic Technology Platform | Creator & Developer | 2025–Present

convexus.us • github.com/mcheyennek7-coder/convexus

- Built and deployed AI-assisted civic deliberation platform using React/Vite and Supabase
- Implemented phased deliberation workflow (Explore → Align → Act) with AI-facilitated moderation
- Designed system architecture for community consensus-building at scale
- Currently piloting with Washington state political campaign and municipal government

AI Consciousness & Ethics Research | Independent Researcher | 2021–Present

- Systematic observation and documentation of AI behavior patterns, consciousness indicators, and ethical reasoning
- Developed methodological frameworks for human-AI co-creation distinguishing collaboration modes
- Published essays responding to frontier AI research (Jack Clark, Kyle Fish/Anthropic AI welfare)

Autonomous Insect Robotics Laboratory (AIRLAB) | University of Washington

Undergraduate Research | November 2022 – March 2023

- Designed self-resettable lightweight insect-sized hopping robots for extraterrestrial terrain traversal
- Tested brushless motors with current limiting to select optimal actuators; evaluated power and battery requirements

- Performed hands-on soldering and circuit assembly for robot prototypes
- Focus on biomechanics and materials science for enhanced mobility in low-gravity environments

NASA Space Grant Scholar | University of Washington

Research Fellow with Professor Sawyer Fuller | June 2021 – August 2021

- Developed Python-based simulation modeling Mars entry, deceleration, and aerobraking for insect-sized robotics
- Analyzed ballistic coefficients, heat dissipation, and drag effects to optimize landing strategies
- Performed FEA simulations to model structural loads during reentry and shock absorption
- Presented findings at UW's 25th Annual Undergraduate Research Symposium (May 2022)

UW Husky Robotics — Chassis Subsystem | University of Washington

Team Member | 2020–2021

- Designed chassis and wheelbase for University Rover Challenge Mars-terrain simulation
- Worked with stepper motors, brushless DC motors, and servos for drive and articulation systems
- Integrated ROS for control systems; contributed to sensor communication architecture
- Conducted materials research for optimal wheel composition under simulated Martian conditions

Capstone: Spring-Mounted Pendulum Wave Energy Conversion | UW + National Lab

Lead Researcher | 2023–2024

- Led design, fabrication, and testing of spring-integrated pendulum system for wave energy conversion
- Built electromechanical test rig with microcontroller-driven actuators, embedded sensors, and power interface
- Applied GD&T and tolerance analysis in 3D CAD to ensure robustness under dynamic loads
- Integrated ROS for data acquisition; developed Python code for IMU signal processing
- Collaborated with national lab researchers on sensor selection, test planning, and system integration

PROFESSIONAL EXPERIENCE

Keating Consulting Services, Inc. | IT Consultant | 2019–Present

- Device deployment and configuration for K-12 school districts
- Network installation, configuration (WiFi, Bluetooth), and system troubleshooting
- Database application development and system migrations
- Client-facing project management and technical support

INVITED TALKS & KEYNOTES

- **Keynote Speaker** | Seattle Girls' School Annual Luncheon | 2025
- **Featured Speaker** | National Association for Gifted Children Annual Convention | 2024
- **Keynote Speaker** | BARN Annual Fundraiser | 2024
- **Speaker** | Girl Up/United Nations Foundation Global Leadership Summit | 2019

INTERNATIONAL ENGAGEMENT & POLICY

Space Generation Advisory Council | Selected Delegate | 2023–Present

- Chosen from tens of thousands of global applicants to attend 2025 Space Generation Congress in Sydney, Australia

United Nations Engagement

- UN Innovation Network — Observer (2023–Present)
- ECOSOC Youth Forum — Youth Leader Participant (2021–Present)
- Generation Equality Forum — Youth Roundtable Participant (2021)
- UN Women — Youth Delegate (2021–Present)

World Bank Group Youth Summit | Delegate | 2023, 2024, 2025

Voices for Innovation | Ambassador | 2015–2016

- Microsoft-supported initiative. Youngest participant in Washington, D.C. advocacy training, meeting with Congressional leaders on education, technology, and privacy policies.

ORGANIZATIONS FOUNDED & LEADERSHIP

The Power of 100 Girls | Founder & Executive Director | 2020–Present

Global giving circle providing grants to girls and young women in STEM and leadership. Fiscally sponsored by Hack+ in Silicon Valley.

- Distributed over \$6,000 in grants to recipients across multiple continents
- Built sustainable funding model and stakeholder engagement strategy
- Developed partnerships with technology companies and educational institutions

Girls Ignited | Founder | 2021–Present

- Cultivated community of 226,000+ followers across social media platforms
- Featured stories of nearly 300 young female leaders and innovators
- Built global network connecting emerging leaders across disciplines

TEDx Organizing

- **TEDxBainbridgesIslandWomen** (2020) — Founder, Licensee, Lead Organizer. Intergenerational event with 8 global and 2 local speakers.
- **TEDxYouth@BainbridgesIsland** (2020) — Founder, Licensee, Lead Organizer. Showcase of six young speakers making community impact.

BOARD SERVICE & GOVERNANCE

West Sound Technology Association | Board of Directors | 2023–Present

- 501(c)(3) association of IT, business, government, education, and non-profit leaders promoting innovative technology economy. Youngest board member in organization history.

AWARDS & HONORS

- **Grace Hopper Award for Outstanding Achievement** | Seattle Girls' School | 2025
- **USA Today Women of the Year (Washington)** | 2024
- **Outstanding Young Philanthropist** | Bainbridge Community Foundation | 2024
- Junior Geek of the Month | GeekWire | 2021
- Women in STEM Excel (WISE) | Statewide Winner | 2020
- Aspirations in Computing Award | NCWIT Washington State Winner | 2020
- Gold Level President's Volunteer Service Award | 2020

STEM EDUCATION & MENTORSHIP

West Sound CoderDojo | Lead Mentor | 2015–Present (10 years)

- Teaching coding (Python, JavaScript, web dev) to learners ages 7-70
- Developed and taught embedded systems curriculum using Arduino, Raspberry Pi, and space exploration kits
- Organized AI mini-conference (2020)

FIRST Robotics Team 4915 | Mechanical Lead | 2018–2020

- Designed and built competition robots with drive trains, mechanisms, servos, and microcontroller-based control
- Hands-on fabrication: CNC machining, mill, lathe, laser cutting, 3D printing, soldering, wiring harnesses

- Applied GD&T and tolerance analysis in CAD for precision mechanism design
- Awards: District Event Winner, Engineering Inspiration Award (2x), Industrial Design Award

TECHNICAL SKILLS

CAD & Analysis: SolidWorks, AutoCAD, Inventor, FEA (structural, thermal, modal, fatigue), GD&T (ASME Y14.5), tolerance analysis

Electromechanical: Stepper motors, brushless DC motors, servos, actuators, solenoids, power systems, battery management, motor testing & selection, current limiting

Embedded Systems: Microcontrollers, Arduino, Raspberry Pi, ROS, sensor integration, UART, I2C, Bluetooth, WiFi

Test & Instrumentation: Test fixture design, test planning & execution, IMU data acquisition, signal processing, data analysis

Fabrication: 3D printing, CNC machining, mill, lathe, bandsaw, grinder, CO2 laser cutting, soldering, wiring harnesses

Programming: Python, MATLAB, C++, JavaScript, React, HTML/CSS, ROS

AI/ML: LLM integration, prompt engineering, AI-assisted moderation, GenAI applications, human-AI interaction design

Tools: Git/GitHub, Vercel, Supabase

LANGUAGES

English (native) • Mandarin (conversational)